🎓 Final Project Report: Digital Literacy & Professional Development

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🚀 Project Overview

This portfolio is a collection of tasks I completed to build my identity as a responsible and technically sound developer. It covers everything from the "raw math" of Machine Learning to the "soft skills" of professional communication and cybersecurity.

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1. Technical Core: Student Marks Predictor

My main coding project is a **Multiple Linear Regression** model built from scratch.

- **The Goal:** Predict a student's marks based on study hours, sleep, and attendance.
- **The Challenge:** I refused to use libraries like scikit-learn. I manually coded the **Stochastic Gradient Descent** loop to understand how weights and biases actually update.
- **Outcome:** I learned that small changes in the "learning rate" (I used 0.0001) can make the difference between a working model and an "exploding gradient" error.

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2. Digital Literacy Awareness (Task 1 & 3B)

I started by researching what it actually means to be digitally literate in 2026.

- **Infographic:** I designed a visual guide covering essential student tools like Google Calendar and Evernote.
- **Data Collection:** I created a **Google Form Quiz** to test awareness. The results showed a 100% success rate in identifying safe passwords, but also confirmed that tools like **ChatGPT** are now central to student life.

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3. Professional Identity (Task 2 & 3A)

I established my presence on the "Big Three" platforms for developers: **LinkedIn, GitHub, and Stack Overflow**.

- **GitHub:** I set up a specialized README to showcase my Java and Python skills.
- **HackerRank:** I began my competitive programming journey by solving the **Java If-Else** challenge, earning 10 points toward my Java badge.

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4. Professional Communication (Task 4)

Communication is just as important as code. I practiced three specific types of formal interaction:

- **Emails:** I drafted a respectful assignment extension request for my **MAT-2005** course and a focused summer internship application targeting **AI-related roles**.
- **Collaboration:** I wrote a report on why group projects fail (usually due to vague messages like "I'll do the report") and how shared task lists can fix it.
- **Checklist:** I made a quick "Do's and Don'ts" guide for social media to keep my professional image clean.

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5. Cybersecurity & Case Study (Task 5)

The final phase was about staying safe in a high-risk digital world.

- **The Prevention Checklist:** I focused on **UPI/Financial safety**, specifically warning against sharing OTPs or scanning random QR codes. I also documented the **National Cyber Crime Helpline (1930)** for emergencies.
- **Cyberbullying Case Study:** I analyzed the heavy impact of online harassment. With **59% of teens** experiencing this, I looked at how rude comments escalate into harmful rumors.
- **The Solution:** Promoting respectful communication and using privacy settings are the best ways to reduce these risks.

💡 Final Reflections

Looking back at these five tasks, I realize that being a "CSE Core" student isn't just about writing code that runs. It’s about:

1. **Safety:** Protecting my data and my finances.
2. **Ethics:** Being a positive influence in online communities.
3. **Clarity:** Communicating my ideas clearly to professors and teammates.

This portfolio is just the beginning of my four-year journey at VIT Bhopal.